

Bacterial 4-Hydroxybutyrate Oxidation to Succinate: A Two-Reaction Module and Its Variants

A commissioned review-style synthesis for a molecular-biology audience. Search date: 2026-08-11. Evidence base: primary literature retrieved via PubMed (PMIDs cited inline).

1. Executive summary

Bacterial oxidation of 4-hydroxybutyrate (4HB) to succinate is a compact, reusable two-reaction module: (i) **4-hydroxybutyrate dehydrogenase** oxidizes the terminal alcohol of 4HB to the corresponding aldehyde, **succinate semialdehyde (SSA)**, reducing NAD^+ ; then (ii) a **succinate-semialdehyde dehydrogenase (SSADH)** oxidizes SSA to **succinate**, reducing NAD^+ or NADP^+ . The net effect is conversion of a small, reduced ω -hydroxy acid into a canonical TCA-cycle intermediate, exporting two pairs of reducing equivalents.

The module's defining, committed reaction is the first one — the 4HB dehydrogenase step — because SSA is a hub metabolite that many pathways produce and consume. The second reaction is a **shared downstream node**: the same SSADH activity terminates GABA catabolism, putrescine catabolism, and (in some bacteria) aromatic hydroxyphenylacetate degradation.

Two facts organize almost everything else. First, the **two enzymes belong to unrelated superfamilies** — the 4HB dehydrogenase is an iron-containing "Group III" alcohol dehydrogenase, whereas SSADH is an aldehyde-dehydrogenase (ALDH)-superfamily thiol enzyme. Second, the **SSADH step exists as two cofactor variants** that recur across bacteria: an NADP^+ -specialist (GabD-type) and an NAD^+ -preferring, cofactor-promiscuous form (Sad/YneI-type). Both variants, and the overall module, are frequently confused with three neighbouring systems that share metabolites but run different chemistry or the opposite direction: the mammalian mitochondrial GABA-degradation step, the reductive succinate→4HB→butyrate routes of clostridia, and the archaeal 3-hydroxypropionate/4-hydroxybutyrate CO_2 -fixation cycle. Keeping these separate is the central conceptual task of this review.

2. Definition and biological boundaries

2.1 What is inside the system

The core comprises exactly two catalytic activities acting on free (non-CoA) substrates:

- **4-hydroxybutyrate dehydrogenase** (EC 1.1.1.61): $4HB + NAD^+ \rightleftharpoons SSA + NADH + H^+$.
- **Succinate-semialdehyde dehydrogenase** (EC 1.2.1.16 / 1.2.1.79 / 1.2.1.24): $SSA + NAD(P)^+ + H_2O \rightarrow \text{succinate} + NAD(P)H + H^+$.

The intermediate SSA is the obligatory hand-off between the two steps (Finding: "Gbd-produced succinate semialdehyde is oxidized by the selected SAD variant").

2.2 What is explicitly outside

Per the working scope, the following are **excluded** and should be treated separately:

- **Uptake of 4HB / GHB** and transporters (e.g., the GABA permease GabP;

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- **Upstream production of 4HB** (from GABA via SSA reduction, from 1,4-butanediol, γ -butyrolactone, or succinate reduction).
- **The GABA shunt proper** (glutamate $\rightarrow$ GABA $\rightarrow$ SSA), which supplies SSA but is not 4HB oxidation.
- **Downstream TCA-cycle oxidation** of the succinate product.

2.3 Neighbouring systems commonly conflated with it

Three "look-alike" systems share metabolites but are mechanistically or directionally distinct:

1. **The mammalian mitochondrial GABA-degradation step.** Human SSADH performs the identical SSA $\rightarrow$ succinate reaction, but in the mitochondrial matrix, is NAD^+ -specific, and carries a redox-sensing regulatory device absent from the bacterial enzymes (§5.3;

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Extrapolating human enzymology to bacteria is a recurrent error.

2. **Reductive succinate→4HB routes.** In *Clostridium kluyveri* the very genes that encode this module (sucD, 4hbD) operate physiologically in the **reductive** direction, converting succinate (via succinyl-CoA and SSA) to 4HB and onward to 4-hydroxybutyryl-CoA/butyrate

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The enzymes are the same chemistry run backwards; the *direction*, not the catalyst, distinguishes catabolic 4HB oxidation from anabolic 4HB formation.

3. **The archaeal 3-hydroxypropionate/4-hydroxybutyrate (HP/HB) CO₂-fixation cycle.** In ammonia-oxidizing Thaumarchaeota and thermoacidophilic Crenarchaeota, 4HB is a key **CoA-thioester** intermediate carrying carbon *from* succinyl-CoA *to* two acetyl-CoA, using a radical **4-hydroxybutyryl-CoA dehydratase** — not a free-acid 4HB dehydrogenase + SSADH

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This is reductive/assimilatory and does not produce succinate. It is the single most important boundary case (Finding #6).

A fourth, subtler confounder: **SSA reductases / GHB-forming enzymes** (e.g., *Geobacter* GsSSAR/GmSSAR, *Gluconobacter* Gox1801, *E. coli* YihU) run the *reverse* of step 1, reducing SSA to GHB/4HB, and belong to the **β-hydroxyacid dehydrogenase** family, not the Group III ADH family

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Annotating these as "4HB dehydrogenases" without specifying direction and family is a common source of literature confusion.

2.4 Competing definitions

The literature uses "4-hydroxybutyrate dehydrogenase," "γ-hydroxybutyrate (GHB) dehydrogenase," and "succinic semialdehyde reductase" for enzymes that interconvert 4HB/ GHB and SSA — sometimes for the *same* protein assayed in opposite directions

([P 31981617](#)

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For this review, the system is defined by **net oxidative flux 4HB → succinate**, regardless of which enzyme name a given paper used.

3. Mechanistic overview

3.1 The best current model of the sequence of events

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4-hydroxybutyrate
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  | (1) 4-hydroxybutyrate dehydrogenase [Group III Fe-ADH]
  |   NAD+ → NADH
  ▼
succinate semialdehyde (SSA; reactive aldehyde hub)
  |
  | (2) succinate-semialdehyde dehydrogenase [ALDH superfamily]
  |   NAD+ or NADP+ → NAD(P)H ; + H2O
  ▼
succinate → (exits module into the TCA cycle)

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- **Step 1 is obligatory and committed** for 4HB oxidation. It is an alcohol→aldehyde oxidation catalysed by a Group III (iron-containing) alcohol dehydrogenase; it is freely reversible and its assayed direction depends strongly on pH (in *C. kluyveri*, optimum pH ≈ 9.4 for 4HB oxidation vs ≈ 6.1 for SSA reduction;

([P 7606170](#)

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- **Step 2 is obligatory but shared.** SSA oxidation by an ALDH-fold enzyme is essentially irreversible under physiological conditions (aldehyde → carboxylate), which pulls flux forward and detoxifies the reactive aldehyde. The cofactor (NAD⁺ vs NADP⁺) is *conditional* on which SSADH variant is expressed (§4.2).

- **Accessory / conditional elements:** substrate uptake, the transcriptional programme selecting the SSADH isozyme, and cofactor re-oxidation are all outside the two catalytic steps but determine whether the module carries flux.

That the module can carry *net oxidative, growth-supporting* flux is directly established in aerobes: wild-type *Alcaligenes eutrophus* (*Ralstonia/Cupriavidus necator*) H16 cannot grow on 4HB, yet spontaneous mutants (e.g. SK4040) grow on 4HB with doubling times ~3 h, and a cloned 10-kbp locus restores 4HB growth to the wild type (P 7851418).

The 1,4-butanediol→4HB→succinate catabolic route similarly supports growth and underpins plastic bio-upcycling in *Pseudomonas* (P 32256468).

A recurring theme is that this oxidative capacity is often **cryptic in wild-type strains and gated by regulation** (§7, point 1).

3.2 Why the order is fixed

The order is thermodynamically and chemically enforced: SSA is the unique product of step 1 and the unique substrate of step 2. There is no plausible route from 4HB to succinate that bypasses the aldehyde — a direct 4-electron oxidation of a primary alcohol to a carboxylate without a free or enzyme-bound aldehyde intermediate is not part of this system. The reactive aldehyde intermediate is the reason the two activities are typically co-regulated and, in anaerobes, physically clustered (P 8550525):

channelling or rapid consumption of SSA limits its toxicity.

3.3 Catalytic chemistry of the two steps

- **4HB dehydrogenase (Group III Fe-ADH).** Sequence analysis places 4HbD/SucD in the class III alcohol/aldehyde dehydrogenase family, related to the bifunctional AdhE (aad) proteins of *E. coli* and *C. acetobutylicum* (P 8550525).

The purified *C. kluyveri* enzyme is a homodimer (~42 kDa subunit) containing metal cofactors (2 Cu + 1 Fe per monomer) and is inactivated by O₂ (t_{1/2} ≈ 4.5 min), consistent with

a metal-dependent hydride-transfer mechanism and an anaerobic physiological context

([P 7606170](#)).

- **SSADH (ALDH superfamily).** SSADH uses the canonical ALDH mechanism: a catalytic cysteine attacks the aldehyde carbonyl to form a thiohemiacetal, hydride is transferred to the nicotinamide C4 to give a thioacylenzyme, and hydrolysis releases the carboxylic acid. In the NAD(P)-dependent YneI (Sad-type) enzyme from *Salmonella*, the catalytic Cys268 and conserved Trp136, Glu365 and Asp426 are essential, and the NAD⁺ cofactor sits in a long channel with its nicotinamide near Cys268

([P 23229889](#)).

The enzyme is a two-domain protein with the active site at the interdomain interface, the standard ALDH architecture

([P 23229889](#)).

([P 20174634](#)).

4. Major molecular players and active assemblies

4.1 Step 1 — 4-hydroxybutyrate dehydrogenase family

Property	Representative: <i>C. kluyveri</i> 4HbD	Notes
EC / reaction	1.1.1.61; 4HB + NAD ⁺ ⇌ SSA + NADH	Reversible; direction set by pH & mass action
Family	Group III (Fe-containing) ADH; class III ADH; AdhE-related	Distinct superfamily from SSADH (P 8550525)
Oligomer / cofactors	Homodimer; 2 Cu + 1 Fe per monomer	Metal-dependent (P 7606170)
Cofactor	NAD(H)	Some homologues assayed as NADPH-dependent SSA reductases (P 25425279)
O ₂ sensitivity	Inactivated by O ₂ (t _{1/2} ≈ 4.5 min)	Constrains clostridial enzymes to anaerobiosis (P 7606170)

The kinetics of "γ-hydroxybutyrate dehydrogenase" enzymes in this Group III Fe-ADH family have been examined explicitly ([P 31981617](#)),

underscoring that the *same* catalytic family provides both the oxidative (4HB→SSA) and, when run in reverse, the reductive (SSA→GHB/4HB) activities.

4.2 Step 2 — succinate-semialdehyde dehydrogenase variants

The **defining axis of variation** in the system is the SSADH cofactor/enzyme family (Finding #2). Two paralogous types recur:

Feature	GabD-type (NADP ⁺ -linked)	Sad / YneI-type (NAD-preferring)
Cofactor	NADP ⁺ -specific	NAD ⁺ and NADP ⁺ ; ~10× higher affinity for NAD ⁺ (P 23229889)
Structural basis of preference	3-residue deletion enlarges/reshapes cofactor pocket to admit the 2'-phosphate; contrast NAD ⁺ -using human enzyme (P 20174634)	Conserved Lys160 contributes to NAD ⁺ preference (P 23229889)
Kinetics	High activity/affinity for SSA	Substrate inhibition above ~0.1 mM SSA (P 23229889)
Regulation	gab operon; nitrogen/Nac/σS control; not putrescine-induced	Putrescine-inducible (YneI); repressed by succinate & low aeration (P 20639325)
Representative structures	<i>E. coli</i> GabD (P 20174634) (P 20060383)	<i>Salmonella</i> YneI (P 23229889)

This dual architecture is **not an *E. coli* idiosyncrasy**: *Klebsiella pneumoniae* independently carries a large NADP⁺-specific SSADH and a smaller NAD-linked form (the latter co-induced with hydroxyphenylacetate catabolism and by GABA/SSA), and a mutant lacking the NAD-linked enzyme retained the NADP⁺-specific one ([P 2647149](#)),

Finding #5). Many genomes therefore encode **≥2 SSADHs**, an NADP⁺-specialist and an NAD-preferring generalist, which are differentially deployed.

4.3 Regulatory and accessory assemblies (context, not core)

- **gabDTPC operon (*E. coli*):** SSADH (GabD), GABA-aminotransferase (GabT), permease (GabP), regulator (GabC/ygaE, a repressor); induced under nitrogen limitation through the Ntr/Nac system, with σ S enabling GABA use as a carbon source

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- **Puu (putrescine) route:** PuuE (a second GABA-aminotransferase) + YneI form a putrescine-inducible SSA-generating/oxidizing branch; post-transcriptionally tuned by the sRNA Spot 42 acting on puuE

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- **GabR (*Bacillus subtilis*):** a PLP-dependent MocR/GabR-family transcriptional activator that switches on gabT and gabD in response to GABA; structurally an HTH DNA-binding domain fused to an aminotransferase-like PLP domain

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 and a candidate antimicrobial target with no eukaryotic homologue
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5. Evolutionary and cell-biological variation

5.1 Lineage variation

- **Anaerobic Firmicutes/Bacteroidetes (e.g., *C. kluyveri*, *C. aminobutyricum*, *Porphyromonas*).** Possess the clustered, O₂-sensitive Group III 4HB dehydrogenase and CoA-dependent enzymes; physiologically these organisms often run 4HB *formation* and further reduce it to butyrate via 4-hydroxybutyryl-CoA

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The oxidative direction is chemically available but not the dominant physiological flux in these strains.

- **Enterobacteria (*E. coli*, *Salmonella*, *Klebsiella*).** Emphasis is on the SSADH node fed by the GABA shunt and putrescine catabolism; two SSADH isozymes with distinct cofactor use and regulation

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- **Aerobic Alphaproteobacteria (*Gluconobacter*).** Provide efficient SSA reductases (NAD(P)H) of the β -hydroxyacid dehydrogenase family

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— the reverse activity — illustrating that "4HB $\leftrightarrow$ SSA" interconversion is distributed across multiple, unrelated enzyme families depending on lineage and physiological need.

5.2 Physiological-state and compartment variation

Isozyme choice tracks physiological programme rather than cell type (bacteria being unicellular): the NADP⁺-specialist GabD serves nitrogen-scavenging/GABA catabolism, while NAD-preferring YneI is switched on by putrescine and repressed by succinate and low aeration

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Compartmentally, the bacterial enzymes are **cytosolic**, in contrast to the **mitochondrial-matrix** localization of the mammalian orthologue

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— a difference with regulatory consequences (§5.3).

5.3 Conservation and origin

- **Deep antiquity of both catalytic scaffolds.** The ALDH superfamily (SSADH) and the Group III Fe-ADH family (4HB dehydrogenase) are both ancient, broadly distributed folds present across Bacteria, Archaea and Eukarya; the catalytic machinery (ALDH catalytic Cys/Glu; metal-dependent Group III ADH) is conserved from bacteria to humans

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- **Ancestral role best read from single-copy, catabolic representatives.** Where the SSADH family has expanded to paralog pairs, the **NAD-preferring Sad/YneI-type** enzyme is the better proxy for the ancestral, cofactor-promiscuous SSA-oxidizing role; the **NADP⁺-specific GabD-type** appears to be a derived specialization achieved by a *small* active-site change (a ~3-residue deletion that admits the 2'-phosphate;

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Cofactor specialization is thus an easily tuned, likely recurrent elaboration rather than a deep dichotomy.

- **Later, lineage-specific elaboration in mammals.** The mammalian mitochondrial SSADH acquired a **redox switch** — a reversible Cys340–Cys342 disulfide on a dynamic catalytic loop that responds to ROS/redox status

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E. coli SSADH structurally lacks this device (a rigidified catalytic loop, no disulfide-mediated change), i.e., the bacterial enzymes represent the ancestral, non-redox-gated state and the switch is a eukaryotic add-on

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Finding #4).

6. Constraints, dependencies, and failure modes

- **Obligatory order.** 4HB must be oxidized to SSA before SSA can be oxidized to succinate; there is no validated aldehyde-bypass (§3.2).
- **Reactive-intermediate constraint.** SSA is a toxic reactive aldehyde; cells that generate it must consume it. Loss of SSADH is lethal/growth-limiting when SSA accumulates — in *E. coli* both SSADHs guard against "accumulation of toxic levels of succinic semialdehyde"

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and a dehydrogenase-less mutant cannot use GABA

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- **O₂-sensitivity constraint.** The clostridial Group III 4HB dehydrogenase is oxygen-labile

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restricting that particular implementation to anaerobic/microaerobic conditions; aerobes accomplish the alcohol↔aldehyde interconversion with different (β-hydroxyacid dehydrogenase-type) enzymes

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- **Substrate-inhibition constraint.** YneI/Sad is inhibited by SSA above ~0.1 mM

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tuning it to low-SSA, high-throughput operation and shaping which isozyme is useful at high flux.

- **Cofactor/redox constraint.** The GabD (NADP⁺) vs Sad (NAD⁺) choice partitions the electrons between anabolic-reductant (NADPH) and catabolic-reductant (NADH) pools; this dictates how the module integrates with cellular redox balance and is a key lever in metabolic engineering (below).

- **Mutual-exclusivity / directionality.** Oxidative 4HB→succinate and reductive succinate→4HB (or SSA→GHB) are opposing fluxes that cannot proceed simultaneously with net gain; the operative direction is set by cofactor ratios, pH, and mass action

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This is why the same enzymes support poly-4HB, 1,2,4-butanetriol, and PHA biosynthesis when run reductively in engineered strains

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— the catalysts are shared; only the direction differs.

7. Controversies and open questions

1. Is 4HB→succinate a bona fide catabolic pathway, or mostly a reconstructed/engineered flux? *Partially resolved.* Direct, growth-based genetics in *Alcaligenes eutrophus* (*Ralstonia Cupriavidus*) show that oxidative 4HB utilization is real and growth-supporting, but is **cryptic in the wild type and activated by regulatory mutation** — spontaneous mutants gain 4HB growth (~3 h doubling), and a defined 10-kbp locus confers the capacity on the parent

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1,4-butanediol→4HB→succinate likewise supports growth in *Pseudomonas*

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That said, much enzymology still comes from organisms running these reactions in reverse (clostridial 4HB formation;

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or from the SSADH node fed by GABA/putrescine rather than by exogenous 4HB

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The open part of the question is therefore not *whether* catabolic 4HB oxidation occurs but *how widely* it is present as an active (vs latent) capacity, and which 4HB-dehydrogenase family serves it in each lineage.

2. **Enzyme naming vs measured direction.** Many "4HB/GHB dehydrogenases" are characterized as SSA *reductases*

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Because these enzymes are reversible, claims about physiological direction should rest on flux/expression data, not on the enzyme name or an in-vitro assay run in one direction.

3. **Why two SSADH cofactor variants?** The adaptive logic of maintaining an NADP⁺-specialist alongside an NAD-prefering generalist is not settled. Differential regulation

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suggests niche partitioning (nitrogen scavenging vs putrescine catabolism vs redox balancing), but the selective advantage of strict NADP⁺ specificity for GabD remains an open question.

4. **Family assignment of "the" 4HB dehydrogenase.** The oxidative first step is provided by at least two unrelated scaffolds across bacteria — Group III Fe-ADH (clostridial 4HbD) and β -hydroxyacid dehydrogenase-type enzymes (Geobacter/Gluconobacter SSA reductases run in reverse). Which family is the physiologically relevant 4HB oxidant differs by organism and is often assumed rather than demonstrated.

5. **Cross-organism extrapolation.** Human/mammalian SSADH data (mitochondrial, NAD⁺-specific, redox-switched) are frequently imported into bacterial discussions; the redox-switch difference

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vs 19300440) shows this is unsafe. Conclusions should be kept organism-specific.

6. **Structure of the bacterial 4HB dehydrogenase.** High-resolution structures and a detailed catalytic mechanism for the Group III Fe-ADH 4HB dehydrogenase are comparatively sparse relative to the well-crystallized SSADHs, leaving the metal geometry and hydride-transfer details less firmly established.

8. Key references

- Söhling & Gottschalk (1996) Molecular analysis of the anaerobic succinate degradation pathway in *Clostridium kluyveri*.

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— sucD/4hbd/cat1 cluster; class III ADH assignment.

- Wolff & Kenealy (1995) Oxygen-sensitive 4-hydroxybutanoate dehydrogenase from *C. kluyveri*.

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— dimer, 2 Cu + 1 Fe, O₂-sensitivity, direction-dependent pH optima.

- Söhling & Gottschalk (1993) CoA-dependent succinate-semialdehyde dehydrogenase from *C. kluyveri*.

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- Gerhardt et al. (2000) 4-aminobutyrate fermentation and 4-hydroxybutyryl-CoA enzymes; abfD/abfT/abfH; homologues in *C. difficile*, *P. gingivalis*, *Archaeoglobus fulgidus*.

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- Zheng et al. (2013) Structure/biochemistry of NAD(P)⁺-dependent SSADH YneI from *Salmonella*.

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— NAD⁺ preference, Lys160, Cys268, substrate inhibition.

- Langendorf et al. (2010) X-ray structure of *E. coli* GabD SSADH; NADP⁺ interactions.

P 20174634

— 3-residue deletion enables NADP⁺ use.

- Ahn, Kim & Kim (2010) Non-redox-regulated SSADH from *E. coli*.

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— bacterial enzyme lacks the redox switch.

- Kim et al. (2009) Redox-switch modulation of human SSADH (Cys340–Cys342).

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modeling

[P 26422261](#)

- Kurihara et al. (2010) Putrescine-inducible P_{puuE}-Y_{neI} GABA-degradation pathway.

[P 20639325](#)

- Schneider et al. (2002) *E. coli* gabDTPC operon; Nac/Ntr/ σ S control; GabC repressor.

[P 12446648](#)

- Sanchez et al. (1989) Two SSADHs in *Klebsiella pneumoniae* (NADP⁺-specific + NAD-linked).

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- Okuda et al. (2015) *B. subtilis* GabR domain architecture, activates gabT/gabD.

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PLP-analogue targeting

[P 39720892](#)

- Meyer et al. (2015) *Gluconobacter* Gox1801 SSA reductase; comparison to *Geobacter* SSAR and *E. coli* YihU.

[P 25425279](#)

Geobacter β -HAD SSA reductases

[P 24878278](#)

[P 22037946](#)

- Liu et al. (2021) Energetically efficient archaeal 3-HP/4-HB CO₂-fixation cycle.

[P 34290692](#)

- Taxon et al. (2020) Kinetics of γ -hydroxybutyrate (Group III Fe-ADH) dehydrogenase.

[P 31981617](#)

- Valentin et al. (1995) *Alcaligenes eutrophus* growth on 4HB; cryptic locus conferring 4HB utilization.

[P 7851418](#)

— direct catabolic, growth-based evidence.

- Li et al. (2020) 1,4-butanediol metabolism (via 4HB) and bio-upcycling in *Pseudomonas*.

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- Metabolic-engineering context (module run reductively):

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Explicit statements of uncertainty

- Physiological net oxidation of free 4HB to succinate as a growth-supporting catabolic pathway **is directly demonstrated in some aerobes**

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but is often **cryptic/regulated**, and its breadth across bacteria — and which 4HB-dehydrogenase family serves it in each lineage — is not fully mapped.

- Enzyme reversibility means **directionality claims require flux/expression evidence**, not in-vitro assay orientation alone.
- Structural knowledge is **asymmetric**: SSADHs are well characterized crystallographically; the bacterial Group III 4HB dehydrogenase is not.
- Findings from *E. coli/Salmonella/Klebsiella*, clostridia, and mammals should **not be pooled uncritically**; compartment, cofactor, and regulatory features differ.